

Moein Razavi

AI/ML Engineer | Data Scientist | Software Engineer

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Professional Summary

Data scientist and software developer with a Ph.D. in Industrial Engineering and experience at Ford, CereVu Medical, and Texas A&M University. Builds AI applications, forecasting systems, and cloud services that turn complex data into usable decisions and workflows. Develops complete applications spanning data pipelines, machine learning, backend APIs, and mobile and web interfaces.

Professional Experience

Ford Motor Company

Data Scientist | Aug 2023 - Present

- **Utility data automation:** Built an AI-assisted onboarding pipeline that turns inconsistent customer electricity-usage records into validated data for energy analysis. Replaced utility-specific parsing workflows with reusable schema discovery, extraction, and validation using FastAPI, pandas, Gemini, and BigQuery.
- **EV analytics assistant:** Enabled natural-language questions about vehicle programs, utility rates, and technical documentation through a multi-agent application. Combined LangChain routing, BigQuery text-to-SQL, document retrieval, and an answer-checking step to connect responses with relevant data and domain rules.
- **Vehicle-to-grid forecasting and dispatch:** Built software to estimate available fleet energy and select vehicles for grid commitments. Combined transformer forecasts, market prices, and Monte Carlo simulation to account for uncertainty, with load-reallocation logic for vehicles that disconnect.
- **Forecasting models:** Trained Temporal Fusion Transformer and Autoformer models with Optuna tuning on Vertex AI GPUs, comparing vehicle-availability forecasts with a previous-week baseline to support energy planning.
- **Rate recommendation:** Developed an application that uses XGBoost electricity-bill predictions, vehicle telemetry, and household consumption to inform utility-plan recommendations. Built Flask APIs, BigQuery workflows, and a Dash interface with SAML authentication; deployed on Cloud Run through Tekton CI/CD.
- **Energy analytics infrastructure:** Automated preparation of energy-simulator inputs and outputs using PySpark, BigQuery, and Airflow. Built discharge-analysis dashboards and REST APIs to make vehicle energy activity and operational performance available to business teams.
- **Code analysis and cloud cost estimation:** Built a FastAPI tool that analyzes repository code, suggests optimizations, and estimates compute costs using reference pricing. Combined Python AST parsing with LLM analysis and streamed findings to the browser as processing completed.

CereVu Medical, Inc.

Software Developer / Research Engineer (Contract) | Jun 2023 - Sep 2024

- Developed Vitality and Clinical, 2 SwiftUI applications connecting wearable sensors with patient and clinician workflows. Integrated CoreBluetooth and Swift Charts to acquire and display 9 physiological signals.
- Migrated the Clinical app from UIKit to modular SwiftUI components using MVVM-Cordinator; implemented role-based views, sensor calibration, data export, and authentication.
- Built Python services for biosignal streaming and a React/Next.js dashboard backed by MongoDB, supporting data review across mobile and web interfaces and iOS delivery through TestFlight.

Selected Projects

- **Business Action Prioritizer:** Built a Google ADK application that turns business data and a scoring rubric into ranked recommendations with supporting reasoning. Required approval of the analysis plan before execution; used PandasAI for analysis and an MCP tool to publish reports to Cloud Storage.
- **InspecaLytics:** Built a field-inspection platform connecting offline data collection, photo annotation, and automated Word and PowerPoint reports. Developed SwiftUI and web interfaces with a shared Supabase backend, cloud synchronization, and Fastlane/GitHub Actions release automation.
- **Clink Social:** Developed a SwiftUI social application with messaging, location-based discovery, and a personalized feed. Built Firebase data services and TypeScript cloud functions for feed aggregation and notifications, plus QR-based cross-device sign-in using time-limited token exchange.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

- **Wearable stress detection:** Developed and evaluated machine learning models using heart-rate and motion data from 54 students over 40 days, with self-reported stress labels. Analyzed predictive features and published findings on stress detection in everyday settings.
- **MERCURY training acceleration:** Co-authored research on reusing computations for similar inputs during neural-network training, published at IEEE HPCA 2023. The study reported a 1.97x average training speedup across 12 models with accuracy similar to its baseline.
- **OpenSync:** Co-developed an open-source Python platform that synchronizes neuroscience sensors and experiment events through Lab Streaming Layer. Published benchmarks measured below 0.07 ms root-mean-square recording-time lag across 4 device configurations.
- **Computer vision and health applications:** Developed video-based physical-distance and face-mask monitoring, and GlucoseCoach, an iOS application with a Flask backend for personalized glucose monitoring. Co-authored a scoping review of machine learning methods for stress-related mental health research.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023

Master of Engineering, Computer Engineering | Texas A&M University | May 2021

Technical Skills

Languages: Python, SQL, Swift, TypeScript, JavaScript

AI and Data: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Optuna, LangChain, Google ADK, Gemini, RAG, pandas, PySpark, Airflow

Applications: FastAPI, Flask, REST APIs, WebSocket, React, Next.js, SwiftUI, UIKit, CoreBluetooth, Firebase, Supabase, MongoDB

Cloud and Delivery: GCP, Vertex AI, BigQuery, Cloud Run, Docker, Terraform, Tekton, GitHub Actions, Fastlane

Selected Publications

First-author publications: OpenSync, Journal of Neuroscience Methods (2022), doi:10.1016/j.jneumeth.2021.109458; scoping review of machine learning and preprocessing for stress-related mental disorders, JMIR Mental Health (2024), doi:10.2196/53714.